Year 11 Set 2 · Higher · grade 5–6 · Lesson 4

Trigonometry (SOH CAH TOA & exact values)

w/c 16 November 2026 · OnMaths · Created by Mr Ali

Today I will learn to...

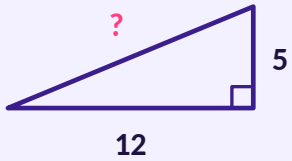
Goals

1. Find missing sides with SOH CAH TOA
2. Find missing angles and use exact trig values

Do now: retrieval practice

4 mins

1 Hypotenuse of a triangle with sides 5 and 12



13

2 Solve $\frac{x}{4} = 7$

x = 28

3 Round 6.4279 to 3 sf

6.43

4 Solve $8 = \frac{20}{x}$

x = 2.5

Objective 1

Find missing sides with SOH CAH TOA

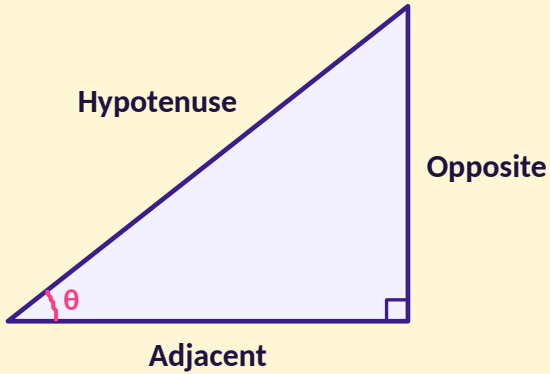
Key idea

Learn

!

Label O, A, H from the angle θ.
Choose sin, cos or tan from the two sides involved.

SOH CAH TOA



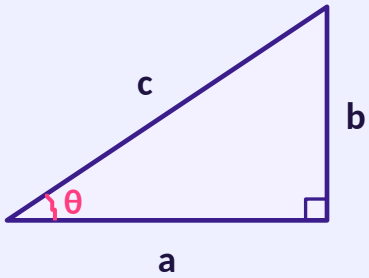
Objective 1: Find missing sides with SOH CAH TOA

Label the sides

We do

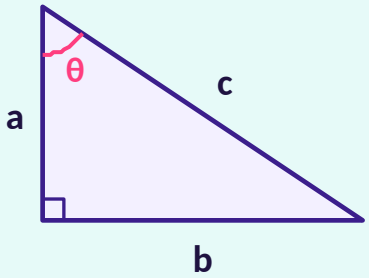
1.

a = A
b = O
c = H



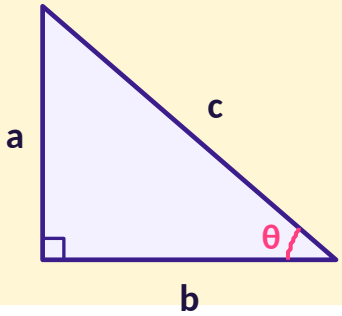
2.

a = A
b = O
c = H



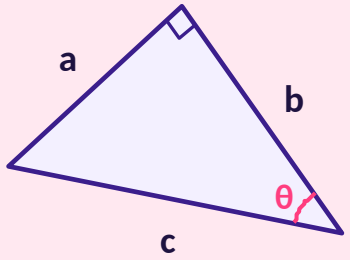
3.

a = O
b = A
c = H



4.

a = O
b = A
c = H

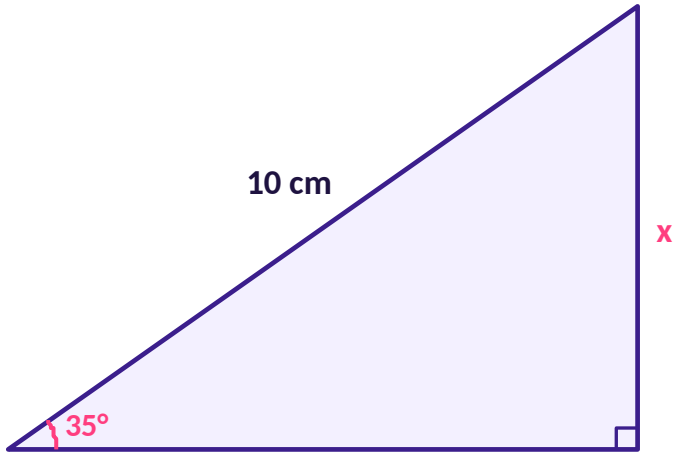


I do: example 1

I do

Angle 35°, hypotenuse 10 cm. Find the opposite side x.

1. Label: H = 10, want O → SOH
2. $\sin 35° = \frac{x}{10}$
3. $x = 10 \times \sin 35°$
4. $x = 5.74$ cm (2 dp)

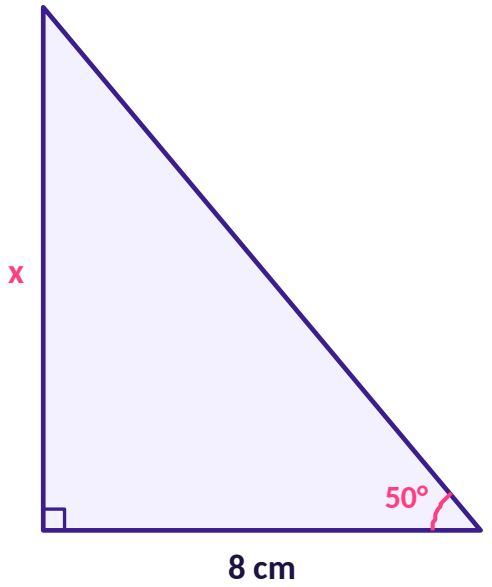


I do: example 2

I do

Angle 50°, adjacent 8 cm. Find the opposite side x.

1. Label: A = 8, want O → TOA
2. $\tan 50° = \frac{x}{8}$
3. $x = 8 \times \tan 50°$
4. x = 9.53 cm (2 dp)

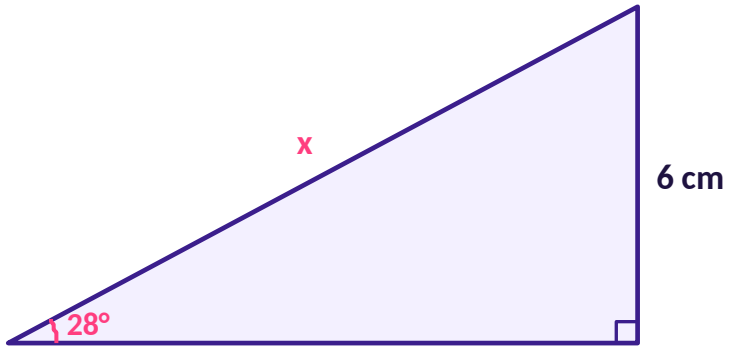


I do: example 3

I do

Angle 28°, opposite 6 cm. Find the hypotenuse x.

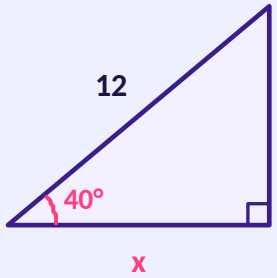
1. Label: O = 6, want H → SOH
2. $\sin 28° = \frac{6}{x}$
3. $x = \frac{6}{\sin 28°}$
4. x = 12.78 cm (2 dp)



Mini whiteboards

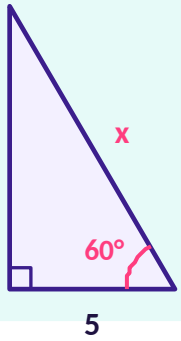
We do

1. Hyp 12, angle 40°. Adjacent (2 dp)?



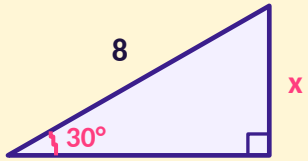
Answer: 9.19

2. Adj 5, angle 60°. Hypotenuse?



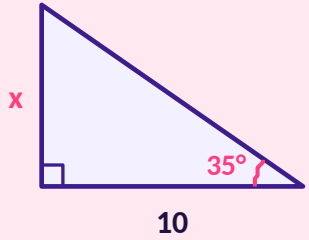
Answer: 10

3. Hyp 8, angle 30°. Opposite?



Answer: 4

4. Adj 10, angle 35°. Opposite (1 dp)?



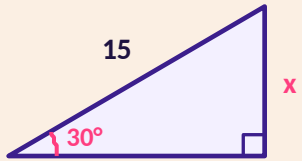
Answer: 7.0

You do: choose your challenge

You do

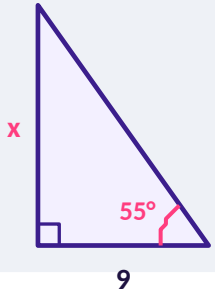
Bronze

Hypotenuse 15 cm, angle 30°. Find the opposite side.



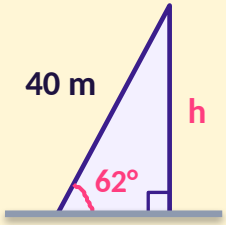
Silver

Adjacent 9 cm, angle 55°. Find the opposite (1 dp).



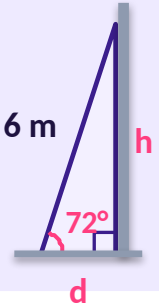
Gold

A 40 m kite string makes 62° with the ground. How high is the kite (1 dp)?



Platinum

A 6 m ladder makes 72° with the ground. How far is its foot from the wall, and how high does it reach (1 dp)?



You do: answers

Check

Bronze

7.5 cm

Silver

12.9 cm

Gold

40 sin 62° = 35.3 m

Platinum

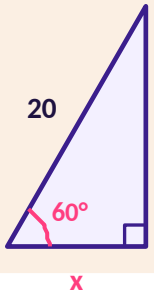
Foot 1.9 m, height 5.7 m

If you finish early

Extra

Bronze

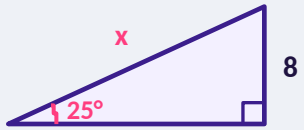
Hypotenuse 20 cm, angle 60°. Find the adjacent side.



10 cm

Silver

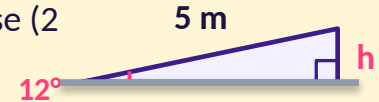
Opposite 8 cm, angle 25°. Find the hypotenuse (1 dp).



$\frac{8}{\sin 25°} = 18.9$ cm

Gold

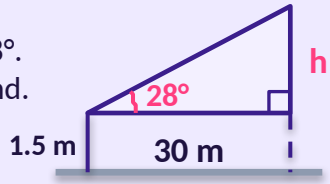
A 5 m ramp makes 12° with the ground. How high does it rise (2 dp)?



5 sin 12° = 1.04 m

Platinum

From 30 m away, the angle of elevation to the top of a tree is 28°. Eye level is 1.5 m above the ground. How tall is the tree (1 dp)?

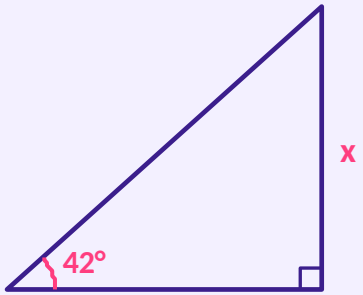


30 tan 28° + 1.5 = 17.5 m

Exam question 1

2 marks

A right-angled triangle has an angle of 42° and adjacent side 7.5 cm. Find the opposite side to 1 dp.



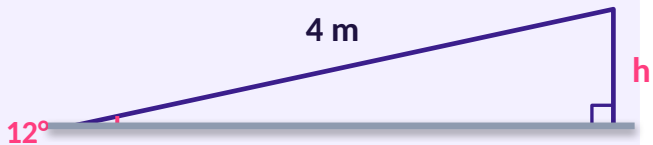
Space for working

Mark scheme: 7.5 × tan 42° = 6.8 cm

Exam question 2

2 marks

A ramp 4 m long makes an angle of 12° with the ground. How high is it? (2 dp)



Space for working

Mark scheme: 4 × sin 12° = 0.83 m

Objective 1 achieved

Find missing sides with SOH CAH TOA



Objective 2

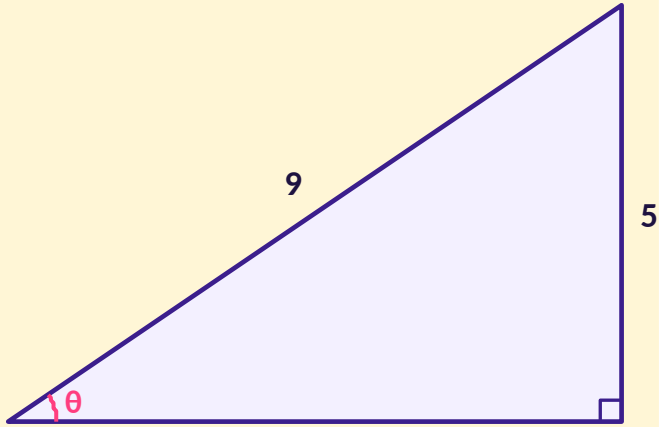
Find missing angles and use exact trig values

✓ Achieved: Find missing sides with SOH CAH TOA

Key idea

Learn

! Use the inverse button: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.
Learn exact values for 0°, 30°, 45°, 60° and 90° for the non-calculator paper.



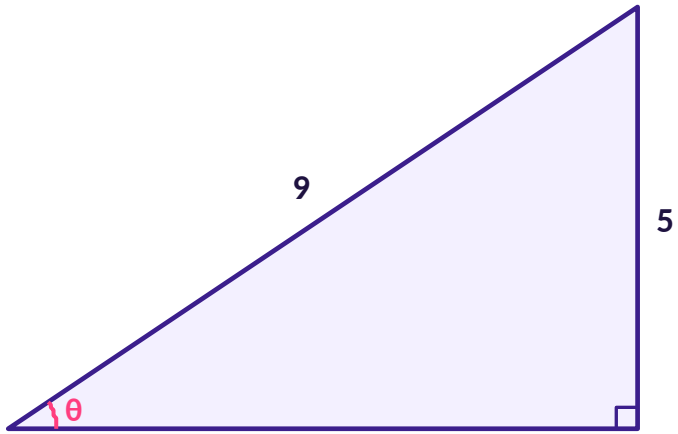
Objective 2: Find missing angles and use exact trig values

I do: example 1

I do

Opposite 5, hypotenuse 9. Find the angle θ.

1. Label: O = 5, H = 9 → SOH
2. $\sin\theta = \frac{5}{9}$
3. $\theta = \sin^{-1}\left(\frac{5}{9}\right)$
4. $\theta = 33.7°$

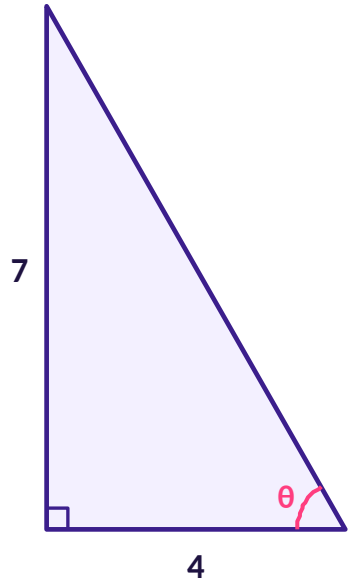


I do: example 2

I do

Adjacent 4, opposite 7. Find the angle θ.

1. Label: O = 7, A = 4 → TOA
2. $\tan\theta = \frac{7}{4}$
3. $\theta = \tan^{-1}\left(\frac{7}{4}\right)$
4. $\theta = 60.3°$

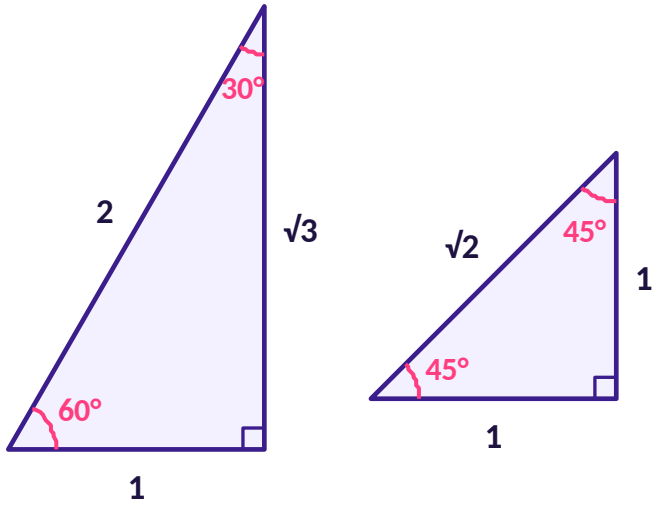


I do: example 3

I do

Exact values to learn (use the two triangles)

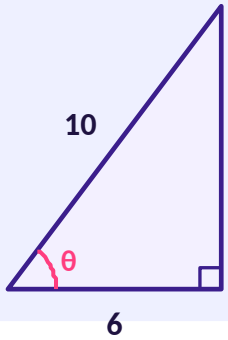
1. $\sin 30° = \cos 60° = \frac{1}{2}$
2. $\sin 60° = \cos 30° = \frac{\sqrt{3}}{2}$
3. $\sin 45° = \cos 45° = \frac{\sqrt{2}}{2}$
4. $\tan 45° = 1, \quad \tan 60° = \sqrt{3}$



Mini whiteboards

We do

1. Adj 6, hyp 10. Angle (1 dp)?

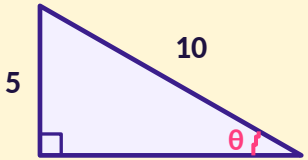


Answer: 53.1°

2. tan 45° = ?

Answer: 1

3. Opp 5, hyp 10. Angle?



Answer: 30°

4. sin 60° exactly?

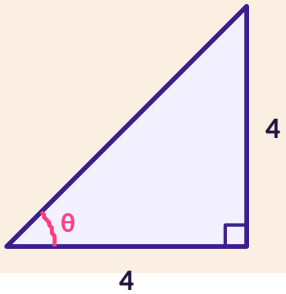
Answer: $\frac{\sqrt{3}}{2}$

You do: choose your challenge

You do

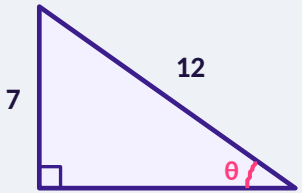
Bronze

Opposite 4, adjacent 4. Find the angle.



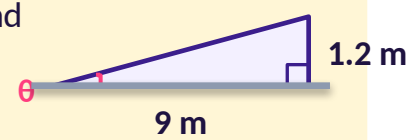
Silver

Opposite 7, hypotenuse 12. Find the angle (1 dp).



Gold

A ramp rises 1.2 m over a horizontal distance of 9 m. Find its angle (1 dp).



Platinum

Without a calculator, find cos 30° × tan 60°.

You do: answers

Check

Bronze

$45°$

Silver

$35.7°$

Gold

$\tan^{-1}\left(\frac{1.2}{9}\right) = 7.6°$

Platinum

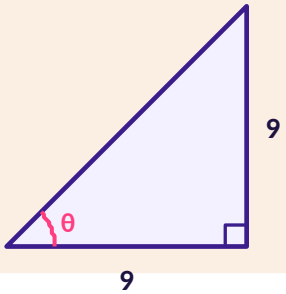
$\frac{\sqrt{3}}{2} \times \sqrt{3} = \frac{3}{2}$

If you finish early

Extra

Bronze

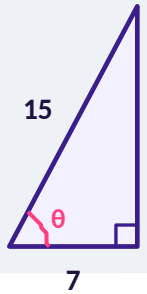
Opposite 9, adjacent 9. Find the angle.



45°

Silver

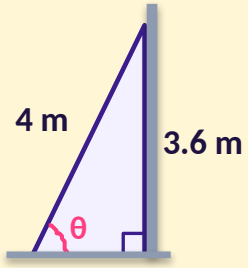
Adjacent 7, hypotenuse 15. Find the angle (1 dp).



$\cos^{-1}\left(\frac{7}{15}\right) = 62.2°$

Gold

A 4 m ladder reaches 3.6 m up a wall. Find its angle with the ground (1 dp).



$\sin^{-1}(0.9) = 64.2°$

Platinum

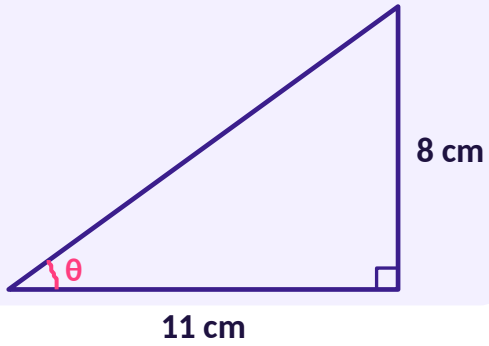
Without a calculator, find $\sin 30° + \cos 60° + \tan 45°$.

$\frac{1}{2} + \frac{1}{2} + 1 = 2$

Exam question 1

2 marks

A right-angled triangle has opposite 8 cm and adjacent 11 cm. Find the angle to 1 dp.



Space for working

Mark scheme: $\tan^{-1}\left(\frac{8}{11}\right) = 36.0°$

Exam question 2

2 marks

Without a calculator, find the value of $\sin 30° + \cos 60°$.

Space for working

Mark scheme: $\frac{1}{2} + \frac{1}{2} = 1$

Objective 2 achieved

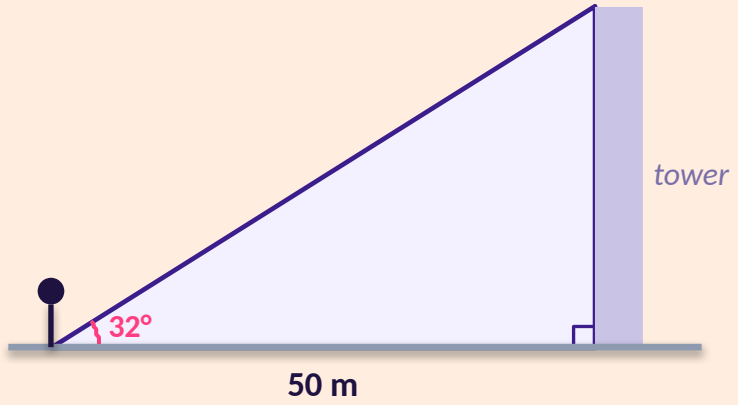
Find missing angles and use exact trig values



Real life

Apply

You stand 50 m from the base of a tower. The angle of elevation to the top is 32°. How tall is the tower? (1 dp)



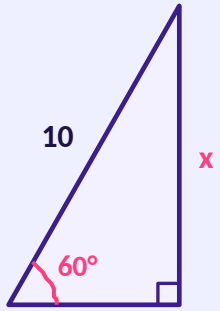
50 × tan 32° = 31.2 m

Extension: stretch yourself

Stretch

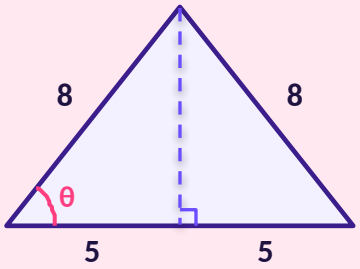
1. **Hypotenuse 10, angle 60°. Find the exact opposite side.**

$10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}$



2. **An isosceles triangle has sides 8, 8 and 10 cm. Find a base angle (1 dp).**

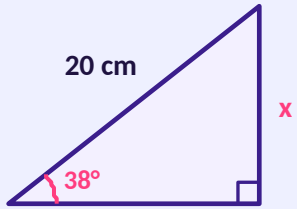
$\cos^{-1}(\frac{5}{8}) = 51.3°$



Exit ticket: exam questions

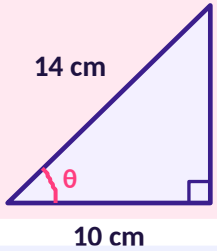
Exit

1. Hypotenuse 20 cm, angle 38°. Find the opposite side to 1 dp.



(2 marks)

2. Adjacent 10 cm, hypotenuse 14 cm. Find the angle to 1 dp.



(2 marks)

3. Without a calculator, find tan 45° + sin 30°.

(2 marks)

Exit ticket: mark scheme

Mark it

Q1 $20 \sin 38° = 12.3$ cm [2]

Q2 $\cos^{-1}\left(\frac{10}{14}\right) = 44.4°$ [2]

Q3 $1 + \frac{1}{2} = 1.5$ [2]

Total: ____ / 6

Today I achieved

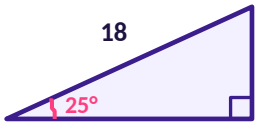
- ✔ **Find missing sides with SOH CAH TOA**
- ✔ **Find missing angles and use exact trig values**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

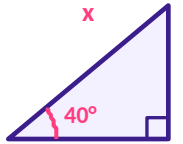
Homework: exam questions

Due next lesson

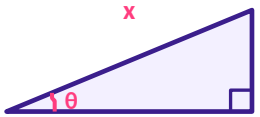
1. Hypotenuse 18 cm, angle 25°. Find the adjacent side to 1 dp. [2]



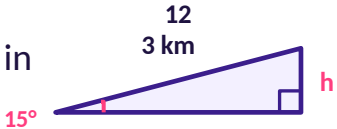
2. Opposite 11 cm, angle 40°. Find the adjacent side to 1 dp. [2]



3. Opposite 5 cm, adjacent 12 cm. Find the angle to 1 dp. [2]



4. A plane flies 3 km along a path that climbs at 15°. How much height does it gain, in metres? [3]



5. Without a calculator, find sin 60° × cos 30°. [2]

Extra practice: Maths Genie → Edexcel question bank → SOHCAHTOA · Exact trig values